High res images!
Paucity of space didn't allow us to put images of these coolers in all their glory. But hey if you want to see what they look like here's the link: dgit.in/CooLerImgs

HOW WE TESTED

Since we got liquid as well as air coolers from many brands, we decided to divide them according to their type for the test. In order to test both the type of coolers, we kept the testbed exactly same.

Test Rig (for Air coolers and liquid coolers):

- Processor: Intel Core i7 3770K;
- Motherboard: ASRock Z77 Extreme 4;
- RAM: Kingston Hyper-X 4GB DDR3 - 1333 MHz;
- Solid State Drive: Intel X25-V 40GB
- Hard Disk Drive: Western Digital Velociraptor 600 GB;
- GPU: Onboard (i.e., Intel HD 4000 series)
- Power Supply Unit: Antec Earthwatts Platinum 650 Watts;
- Cabinet: Cooler Master HAF-XM;
- Cabinet Fans: Front 1 x 200mm (intake), Rear 1 x 140mm (exhaust), Side-panel 1 x 200mm (exhaust);
- Operating System: Windows 8 Enterprise Edition;

Cable management was observed while testing so that cables do not obstruct air flow. All the coolers came with their own thermal paste but in order to maintain a fair playing field, we used Arctic Silver 5 thermal paste for all the coolers while testing. We also applied the same amount of paste for each cooler. In terms of test software, we used RealTemp and SpeedFan to monitor the temperatures of all the CPU cores. For stress testing, Prime95 was used to put the CPU on 100% load and we ran it three times for 10, 10 and 15 minutes for each cooler and noted the temperature readings for all the cores every time we ran the stress test. We then took a note of the max load temperatures of all the cores of the CPU for each cooler and took an average of that and the same was applied all the three times we ran the test. Then we took an average of the three results we got, so that we get a final load temperature. We then left the PC idle for 15 minutes and measured the minimum idle temperature.

It must also be noted that, we took special care to maintain the ambient room temperature at 28°C at all times while testing and to monitor that we used a digital thermometer. For all the coolers, we used the default fans provided with each cooler and installed them according to how the default mounting for each cooler fan should be according to the manufacturers. In liquid coolers for instance, we mounted the fans in such a configuration that they could intake cool air directly from outside the case.

We assigned 75% weightage to the performance, 20% weightage to cooler features and accessories, 5% weightage to build quality. For the performance parameter we included idle performance and load performance. Noise performance was also accounted for. For noise testing, we did a real world test in which we took a note of whether a cooler was audible from outside the fully closed cabinet i.e., if we heard a dull hum or loud humming or whether a cooler was absolutely inaudible or silent. Depending on these three aspects, we gave a score for noise performance for each cooler. We also gave a score to every cooler for ease of installation depending on the number of steps required to mount a particular cooler. For air coolers we also gave points based on RAM clearance an air cooler provides around and above the RAM slots, so if a cooler is very low and touches the RAM or leaves hardly any room for a user to install RAM with tall heat spreaders, then we deducted points for that and if there is sufficient space around and above the RAM slots then it will obviously fare well in terms of the score.

case, meaning you will have to look for the sound of the fan to find it. Another good part about the TX3 Evo is it's RAM clearance. It had the maximum RAM clearance when compared to every other air cooler in this test. And when you speak of installation, the TX3 tops the charts in that aspect too, owing to the fact that it uses the simple push-pin based mounting mechanism which is seen on the Intel stock cooler.

The TX3's bigger cousin, the Cooler Master Hyper T4 belongs to the bigger but slimmer category of single tower air coolers and its mounting is similar to AMD's hook like pressure based mechanism. The included thumbscrews help during installation and since the T4 comes with a single 120mm fan, you can mount your own 120mm fan using the supplied fan mounts. Performance wise the T4 is very similar to the Deepcool GAMMAXX 400 with both of them giving 54°C on full load but the GAMMAXX 400 has an advantage of 1.75°C in idle over the T4.

The GAMMAXX 400 also employs the easy to mount Intel push-pin mechanism. Another added feature of the GAMMAXX 400 is the included blue LED fan which would give your system a nice glow from the inside. Moving on, we have two twins from Cooler Master, which are the Hyper 212 Evo and the Hyper 212X. Based on the immensely popular and older Hyper 212 or Hyper 212 plus from cooler master, the 212 Evo and 212X are identical in terms of almost everything except the new type of vents on each aluminium fin around the heat pipes of the 212X, called as "X-Vents" by Cooler Master which supposedly allows for better heat dissipation. Since the 212 Evo and 212X have so much in common, they are ditto in performance, and we must say that if you are looking for a decent air cooler with good performance, you should think of considering either of the two.

Next up, we have the Deepcool FROSTWIN, a twin tower offering from Deepcool which has a very organised box packaging with every part labelled to aid the user in installing the cooler. Bundled with two pre-installed 120mm fans that are powered from a combined 4-pin fan connector, the heatsink of FROSTWIN is smaller than the included fans but it's design is thought out well and that shows in its performance. The fans on Deepcool FROSTWIN have the new Hydro bearing which allow for better continuous lubrication and silent operation which is seen in its noise performance as FROSTWIN runs very silent even at full loads.

Speaking of other coolers that are bundled with two fans out-of-the-box, we received the Cooler Master Hyper 412 Slim and the Deepcool NEPTWIN, the Cooler Master Hyper 412 Slim is not slim in terms of its heatsink but it has slim fans, probably one of the slimmest we've seen in a long time. In fact if you put these slimmer fans side-by-side and compare